GMU - FACULTY PROFILE

Dr. Arun Kumar K M

Associate Professor
Electronics and Communication Engineering,
GM University
Davanagere - 577006

Faculty
Faculty of Engineering and Technology

School / Program
School of Engineering
B.Tech. in Electronics and Communication Engineering

Contact

Email:
arunkumarkm.fet.se@gmu.ac.in
Phone: 9632682369
Websites/ Academic IDs:
ORCID ID: 0009-0006-4308-7005
Scopus ID:
Researcher ID:
Google Scholar ID:
VCWXjhUAAAAJ
Vidwan ID: 657425

Faculty Introduction

Dr. Arun Kumar K M is an electronics engineering professional with a Ph.D. from VTURRC Belagavi, specializing in vibration signal analysis in dynamic machines. He has over 10 years of teaching experience and 03 years of industry exposure in VLSI design verification, working with companies like Expolog Technologies and Micron Technology.

His expertise includes LPDDR and DDR memory standards, AMBA protocols, serial communication protocol and industry design tools such as Cadence and Mentor Graphics. He has published numerous papers on fault diagnosis and vibration analysis and holds an Indian patent related to vibration suppression systems. His teaching portfolio spans undergraduate and postgraduate subjects, and he actively pursues professional development through workshops and seminar engagement. His career blends academic rigor with practical industry knowledge.

Qualifications

Education	University	Institution	Year	%
Ph. D	VTU, Belgaum	VTU RRC, Belgaum	2019	-
M.Tech (Digital Electronics)	VTU, Belgaum	GM Institute of Technology, Davangere	2012	72.5%
B.E (Electronics &Instrumentation)	VTU, Belgaum	B.I.T, Bangalore	2006	62.5%

Teaching Experience (13 years)

#	Institution	Designation	from	To	Exp.
1	Sri Taralabalu Jagadguru Institute of Technology Ranebennur	Assistant Professor	07-08-2020	10-11-2022	10-years 3-months

Industry Experience

#	Institution	Designation	Clients	from	To	Exp.
1	Expolog Technologies, Davangere	Senior Design Verification Engineer	Micron Technology, Hyderabad	15-11-2022	25-07-2025	2-years 9-months

Research Areas

- Low-power DRAM design and verification, covering LPDDR, DDR3, DDR4, and DDR5 memory.
- Design verification of industry-standard AMBA protocols, including AXI, AHB, and APB bus protocols.
- Serial interface design and verification, focusing on SPI, I2C, and related protocols.

Research Scholars Registered and awarded for Ph.D.

Nil

Research Funding Details

Nil

Patents Details

Faculty Name	Patent Title	Patent Date of Filing with Patent No.	Patent Published Date	Patent Granted Date
Dr. ArunKumar K M	System and method for vibration suppression of different types of structures using multiple sensing techniques using control algorithms	04/06/2021 202111021980	10/11/2021	Not Yet

Publications Details

Scopus Journal Papers	Please see Annexure -I	03
Other Journal Papers	Please see Annexure -II	-
Conference Papers	Please see Annexure-III	24
Books/Book chapters	Please see Annexure-IV	-
	Total Publication	**27**

Workshop/ Training

Organized	-	-
Attended	Please see Annexure-VI	06

List out the Consultancy Projects undertaken

1	NIL

List out the Guided/involved in Innovation workshops/Training

1	NIL

Have you done any contribution to new product development and setting up start-ups?

1	NIL

List of Training Courses Taught with number of students in each course

1	Nil

Professional Recognition

Nil

Professional Membership

1	International Associations of Engineers (IAENG), Life Member (Membership No. LM 136598)

Skill Set

Languages	UVM, System Verilog, VHDL, Verilog.
Synthesis tools	• Cadence (NCSIM), xcelium, xrun, IMC, Vmanager • Mentor Graphics – Questasim • Synopsis (VCS), Verdi • XIlinix - ISE, Xilinx-Vivado
Simulation tools	• Cadence, ModelSim XE III 6.2c, Microwind, ORCAD, SPICE SPACE • Mentor Graphics – Questasim, Synopsis (VCS), Verdi

NPTEL/MOOC Course Completed

Nil

Awards/ Honour's/ Achievements

1	***Best Paper award*** for "Design and verification of AMBA APB protocol using Verilog", National conference on Smart Systems and Emerging Trends in Electronics and Electrical Engineering NCSEEE-2025, GM University, Davangere, Karnataka, India, 21st Aug 2025.
2	Successfully completed with a consolidated score of 76%in "Higher Education Assessment Test" Faculty Quiz Competition, held on 13thSeptember 2025 organized by the GMU Quizers, GM University - (2025)
3	Successfully completed the Faculty Development Programme through LEAP with an average score of 77.3% (Grade B+) held on 30th August 2025 at GM university. - (2025)

Funded Students Projects Guided: PG /UG

Invited Lectures

Present Role and Responsibility

1	Institute Rating and Ranking, GM University

Annexure – I (Scopus Journal Papers)

Academic year – 2017-18 (August to July)		No. of Publication	2
[3]	K.M. Arun Kumar, Dr. G. Arun Kumar Dr.T.C. Manjunath "Stator Winding Fault Diagnosis of IM by Park's Vector Approach and RF based on Vibration Signal", Elsevier's Journal of Advanced Research in Dynamical and Control Systems. ISSN :1943-023X, Special Issue-12, pp. 234-244, 2018. http://www.jardcs.org/archives-special.php?year=2018&issue=12-Special%20Issue&page=4		
[2]	K.M. Arun Kumar, Dr.T.C Manjunath, Dr. G. Arun Kumar "Bearing Fault Diagnosis in IM Using Perk's Vector Approach and J-48 based on Vibration Signal",Elsevier's Journal of Advanced Research in Dynamical and Control Systems, ISSN : 1943-023X,Special Issue-12, pp. 222-233, 2018. http://www.jardcs.org/abstract.php?archiveid=5471		
Academic year – 2015-16 (August to July)		**No. of Publication**	**1**
[1]	Arun kumar K.M., Dr.T.C. Manjunath "A brief review/survey of vibration signal analysis in time domain", SSRG International Journal of Electronics and Communication Engineering (SSRG-IJECE) ISSN: 2348 – 8352 Volume 3 Issue 3 Page No 12-17 March 2016.		

Annexure – II (Other Journal Papers)

Nil

Annexure-III (Conference Papers)

Academic year – 2024-25 (August to July)		No. of Publication	01
[24]	**Arunkumar K.M.**, "Design and verification of AMBA APB protocol using Verilog", National conference on Smart Systems and Emerging Trends in Electronics and Electrical Engineering NCSEEE-2025, ISBN:978-81-991111-0-3 GM University, Davangere, Karnataka, India, 21st Aug 2025.		
Academic year – 2027-18 (August to July)		**No. of Publication**	**09**
[23]	**Arunkumar K.M.**, Dr. T.C.Manjunath, "Experimental evaluation of diagnosis & analysis of bearing faults in induction motors", IEEE Int. Conf. on Intelligent Computing & Control Systems (ICICCS 2018), Vaigai College of Engg., Therkutheru (Post), Melur Taluk, Madurai 625122, Tamil Nadu, India, Paper id 52499A-20170530101416, 14-15 June 2018.		
[22]	**Arunkumar K.M.**, "Experimental Evaluation of Diagnosis & Analysis of Bearing Faults in Induction Motors", International Conference on Intelligent Computing & Control Systems(ICICCS) organized by Vaigai Collage of Engineering, Madurai 14th & 15th June 2018.		
[21]	**Arunkumar K.M.,** Dr. T.C.Manjunath, "Design of controllers for bearing faults", RTEICT 2018 3rd IEEE Int. Conf. on Recent Trends In Electronics, Information & Commun. Tech.,Sri Venkateshwara College of Engg., Sri Venkateshwara College of Engg., Bangalore, Karnataka, India, paper id 1003, 18 May 2018 - 19 May 2018.		
[20]	**Arunkumar K.M.**, Dr. T.C.Manjunath, "Bearing fault diagnosis through vibration signals utilizing J 48 Decision Tree Algorithm", RTEICT 2018 3rd IEEE Int. Conf. on Recent Trends In Electronics, Information & Commun. Tech., Sri Venkateshwara College of Engg., Bangalore, Karnataka, India, paper id 1004, 18 May 2018 - 19 May 2018.		
[19]	**Arunkumar K.M.,** Dr. T.C.Manjunath, "Bearing Faults Diagonsis Through Vibration Signals Utilizing J-48 Decision Tree Algorithm" 3rd IEEE International Conference on Recent Trends in Electronics, Information & communication Technology(RTEICT-2018)organized by sri Venkateshwara College of Engineering, Bangulu IEEE XPLORE ISBN:978-1-5386-2440-1 18th & 19th May 2018.		

[18]	**Arunkumar K.M.,** Dr. T.C.Manjunath, Dr. G. Arun Kumar, "An experimental study (MCSA) of the Broken Rotor Bar Detection of IMs using STFT & RF algo", International Conference on Innovations in Engineering, Technology and Sciences, (SICIETS-2018), Sreenivasa Institute of Technology and Management Studies (SITAMS), Chittoor, AndhraPradesh, India, Paper ID: SICIETS : 049 ECE, 28 - 29 October 2018.
[17]	**Arunkumar K.M**., Dr. T.C.Manjunath, Dr. G. Arun Kumar, "Design & fault identification system of Stator and Rotor Winding Faults for Induction Motors", 6th Nat. Conf. on Emerging Trends in Engg. Res. & Management (NCETERM)-2018, GM Institute of Technology, Davanagere, Karnataka, India, 26-27 Oct. 2018.
[16]	**Arunkumar K.M.,** "Temperature Scanning Controller Design", 6th International Conference on Chip, Circuitry, Current, Coding, Combustion & Composites– i7C-2018, organized by Sri Sairam College of Engineering Bangalure on 28th & 29th November 2018.
[15]	**Arunkumar K.M**., "Modelling of discrete events using Verilog Language", 6th International Conference on Chip, Circuitry, Current, Coding, Combustion & Composites– i7C-2018, organized by Sri Sairam College of Engineering Bangalure on 28th & 29th November 2018.

Academic year – 2016-17 (August to July)	**No. of Publication**	**06**

[14]	Arunkumar K.M., Dr. T.C.Manjunath, "A novel design & development of adaptive noise cancellation in machines using LABVIEW", IEEE International conference on power, control, signals and Instrumentation Engineering (ICPCSI - 2017), ISBN : CFP17M84-PRT 978-1-5386-0813-5, ISSUE-II, Saveetha Engineering College, Thandalam, Chennai,Tamil Nadu, India, pp. 297-302, IEEE Record 42447, XPLORE COMPLIANT CFP17M84-ART ISBN 978-1-5386-0814-2; CDROM CFP17M84CDR ISBN 978-1- 5386-0812-8, PRINT CFP17M84-PRT ISBN 978-1-5386-0813-5, Sep. 21 & 22, 2017.
[13]	**Arunkumar K.M**., Dr. T.C.Manjunath, "Vibration Signal Monitoring Based on Virtual Instrumentation", IEEE International conference on power, control, signals and Instrumentation Engineering (ICPCSI - 2017), ISBN : CFP17M84-PRT 978-1-5386-0813-5, ISSUE-II, Saveetha Engineering College, Thandalam, Chennai, Tamil Nadu, India, pp.303-305, IEEE Record 42447, XPLORE COMPLIANT CFP17M84-ART ISBN 978-1-1355386-0814-2; CDROM CFP17M84-CDR ISBN 978-1-5386-0812-8, PRINT CFP17M84-PRT ISBN 978-1-5386-0813-5, Sep. 21 & 22, 2017.
[12]	**Arunkumar K.M.,** Dr. T.C.Manjunath, "Vibration signal analysis using Time domain and Time Frequency domain: review", IEEE International conference on power, control, signals and Instrumentation Engineering (ICPCSI - 2017), ISBN : CFP17M84-PRT 978-1-5386-0813-5, ISSUE-II, Saveetha Engineering College, Thandalam, Chennai, Tamil Nadu, India, pp. 306-309, IEEE Record 42447, XPLORE COMPLIANT CFP17M84-ART ISBN 978-1-5386-0814-2; CDROM CFP17M84CDR ISBN 978-1-5386-0812-8, PRINT CFP17M84-PRT ISBN 978-1-5386-0813-5, Sep. 21 & 22, 2017.
[11]	**Arunkumar K.M**., Dr. T.C.Manjunath, "Vibration signal monitoring using FFT analysis", 5th Nat. Conf. on Emerging Trends in Engg. Res. & Management (NCETERM)-2017, GM Institute of Technology, Davanagere, Karnataka, India, 8-9 Sept. 2017.
[10]	**Arunkumar K.M**., Dr. T.C.Manjunath, "Vibration signal monitoring of Rotating machines using VI", Int. Conf. on Recent Trends in Engg., Tech. & Sci., ICRETS-2016, ECE Dept., S.T.J.I.T. Ranebennur, Karnataka, India, 12-13 Aug. 2016.
[9]	**Arunkumar K.M**., Dr. T.C.Manjunath, "Vibration signal analysis using time domain, frequency & time frequency domain: A review", Proc. First Nat. Conf. on Recent Trends in Communication & Computing, VTU Sponsored Nat. Conf., ECE Dept., Jain Inst. of Tech., Davanagere, Karnataka, India, Page no. 47, 27 Feb. 2016.

Academic year – 2013-14 (August to July)	**No. of Publication**	**06**

[8]	**Arunkumar K.M.,** Dr. T.C.Manjunath ,"Development of Swarm Intelligence in Robotics Systems", International Conference on "Recent Advances & Challenges in Engineering & Management RACEM-2014", organized by Vidyalankara Institute of Technology Wadala (E), Mumbai on 3rd & 4th April-2014, in Collaboration with IEEE BOMBAY Section.

[7]	**Arunkumar K.M.**, Dr. T.C.Manjunath, "OCR Engine Development in IP arena" 4th national conference on Recent Advances In Power Electronics & Industrial Drives (PEID-2K14) page no.352 355, paper code 2B.001, session 2B, held at INFO Institute of Engineering, Coimbatore on 6th & 7th march, 2014. 5.
[6]	**Arunkumar K.M.**, Dr. T.C.Manjunath ,"Mathematical modeling of a Jacobian Matrix for space applications" 4th national conference on Recent Advances In Power Electronics & Industrial Drives (PEID-2K14) page no 238-241, paper code 1C.013, session 1C, held at INFO Institute of Engineering, Coimbatore on 6th & 7th march,2014.
[5]	**Arunkumar K.M.**, Dr. T.C.Manjunath ,"Microcontroller design & development for a remote control of a communication system" 4th national conference on Recent Advances In Power Electronics & Industrial Drives (PEID-2K14) page no.320-326, paper code 2A.013, session 2A, held at INFO Institute of Engineering, Coimbatore on 6th & 7th march, 2014.
[4]	**Arunkumar K.M.**, Dr. T.C.Manjunath ,"Algorithm development for obstacle avoidance for human beings" 4th national conference on Recent Advances In Power Electronics & Industrial Drives (PEID 2K14) page no. 404-409, paper code 2B.012, session 2B, held at INFO Institute of Engineering, Coimbatore on 6th & 7th march, 2014.
[3]	**Arunkumar K.M.**, Dr. T.C.Manjunath ,"A New Predictive Energy Efficient Bee Routing (PEEBR) Algorithm For Routing In A WSN Network" 43rd ISTE National Annual Convention Empowering Technical Education of Address Sustainability & Global Competitiveness held at Tatyasaheb Kore Institute Of Engineering & Technology, Waranangar, Kolhapur , Maharashtra, ISBN NO 978-93 514315-7 during 19th-21th Dec 2013.

Academic year – 2011-12 (August to July)	**No. of Publication**	**02**

[2]	**Arunkumar K.M.**, "Saccada Planning For Real Time Face Detection And Tracking Using A Single Pan,Tilt, Zoom(PTZ) Camera" International conference on "current trends in Engineering and Management ICCTEM -2012" organized by Vidyavardhaka College of Engineering, Mysore, Karnataka, paper id EC_06,ISBN 978-93-5051-905-9 held during July 12-14,2012.
[1]	**Arunkumar K.M.**, Dr. T.C.Manjunath, "Face Recognition Using Eigen Face and Neural Network Approaches" in Delta 12.0 organized by E&C department GMIT, Davangere during 22-23, April 2012.

Annexure-IV (Books/Book chapters)

NIL

Annexure – V (Workshop /Training /FDPs/ Industry Visit Convened & Organized)

Nil

Annexure – VI (Workshop/Training attended)

6	Workshop on "NI LABVIEW & ARM PROCESSOR" organized by G.M.I.T, Davangere from 27th & 28th February 2015.
5	Workshop on "Time Frequency Analysis & Wavelet Transforms" organized by Department of Aerospace Engineering of I.I.Sc, Bangalore from 14th February 2015.
4	Workshop on "Fundamentals of Lab VIEW & its Applications" organized by RYMEC, Bellary from 25th & 26th July 2014.
3	Workshop on "Signals and Systems" organized by Indian Institute of Technology Kharagpur. National Mission on Education through ICT (MHRD) from 2nd Jan to 11th Jan 2014.
2	Workshop on "Control Systems" organized by Indian Institute of Technology Kharagpur. National Mission on Education through ICT (MHRD) from 2nd to 12th Dec 2014.
1	Workshop on "Latex & its Documentation for Research Scholars" organized by G.M.I.T, Davangere from 9th Nov. 2013.